# PORTER: A Plan-Committed Router for Multi-Core Distributed Quantum Computers

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***Abstract*—A distributed quantum computer runs one circuit across several small cores joined by links that distribute entanglement on demand. Under the usual cost model the compiler's objective is to spend as few Einstein–Podolsky–Rosen (EPR) pairs as possible on gates that straddle cores. Existing routers choose among inter-core actions while simultaneously staging them, so an action's apparent value depends on the layout the compiler happens to occupy. We present PORTER, a compiler that instead scores a complete inter-core *plan* — which primitive fires, which gates it resolves, and which readiness conditions it needs — before any preparation SWAP for it exists, commits to one, and only then prepares it. PORTER supports all three known inter-core primitives, including cat-entangler packets that amortise one pair over a run of remote gates sharing a control, and adds capacity-aware relief, a deterministic initial layout, and a per-circuit choice of teleport-lookahead horizon. On MQT-Bench suites at 25, 64 and 80 logical qubits PORTER uses $59.6\%$, $41.6\%$ and $68.2\%$ fewer EPR pairs in geometric mean than published TELESABRE at three routes each, and completes all twenty-four instances where TELESABRE completes seventeen. Against DSABRE it is competitive at the sizes both target — $1.07\times$ its pooled EPR at $0.58\times$ its compile time under each compiler's own protocol — and pulls ahead as the machine grows, using $0.72\times$ its EPR pairs on a 100–360-qubit scale-up series. Controlled ablations measure what each mechanism contributes. The implementation, harness and raw results are public.**

***Index Terms*—distributed quantum computing, quantum circuit compilation, multi-core quantum architectures, gate teleportation, cat-entangler packets, plan-oriented routing, entanglement cost**

## I. INTRODUCTION

A distributed quantum computer (DQC) runs one circuit across several small cores joined by links that distribute entanglement on demand [1], [2]. Any two-qubit gate whose operands sit in different cores must be paid for with Einstein–Podolsky–Rosen (EPR) pairs, and distributing a pair is far slower and less reliable than any operation inside a core [3], [4]. Pairs consumed is therefore the metric we minimise (Section II-B).

A pair can be spent three ways. Teleporting a *state* (TELEDATA) relocates a logical qubit, making every later gate it meets in its new core local [5]. Teleporting a *gate* (TELEGATE) executes one straddling controlled gate through a cat-entangler, with neither operand moving [6], [7]. Holding that cat-entangled resource open across a run of straddling gates that share a control costs one pair for the whole run — a *packet*, following AutoComm's gate-sequence amortisation [8]. The last two work for any controlled two-qubit gate; our benchmarks use CX throughout (Section II-A). Which primitive is cheapest depends on the circuit structure where it is applied, so none dominates.

The choice must be made before its price is known. Each alternative becomes executable only after its own preparation: the right operand beside the right communication port, that port free, the destination holding a spare slot. A compiler performs that work only for the alternative it selects. Scoring each action together with the preparation it currently needs therefore confuses two things. It ranks candidates by the layout the compiler happens to occupy, not by the progress they buy. And staging a candidate that then loses still perturbs the layout.

PORTER — a *Plan-Oriented Router for Teleportation-Enabled aRchitectures* — separates the two questions. Its unit of choice is a *plan*: a complete specification of one inter-core action and the conditions it needs, omitting the SWAP sequence that would establish them. Plans are compared on the gate progress they buy, one is committed, and only then are preparation SWAPs generated for it. Around this loop PORTER adds two-tier capacity relief, which manufactures free slots when every candidate is blocked, and a single deterministic initial layout in place of the best-of-$N$ search its predecessors run.

**Results.** We evaluate on MQT-Bench [9] suites at 25, 64 and 80 logical qubits, on a 100–360-qubit scale-up series, and on two networks of IBM heavy-hex cores — forty-eight circuit–architecture instances. Against TELESABRE [10], the strongest published router we identified under this cost model, PORTER uses substantially fewer EPR pairs: over the instances both complete it lowers geometric-mean consumption by $59.6\%$, $41.6\%$ and $68.2\%$ on the three suites at three routes each, and it completes all twenty-four, including the seven — among them an 80-qubit, 23,381-CX circuit — on which TELESABRE does not converge.

The second comparison is with DSABRE [11], concurrent work on the same problem and the closest point of reference we have. PORTER is competitive with it at the sizes both target, spending $1.07\times$ its pooled geometric-mean EPR for $0.58\times$ its compile time under each compiler's own protocol, and it pulls ahead as the machine grows: on the 100–360-qubit series it uses $0.72\times$ DSABRE's EPR pairs and wins every instance. Reaching parity with a concurrent compiler from a different design, and then improving on it where the core count rises, is the result we take from this comparison; Section V gives every operating point behind it. The two are close relatives and share four design decisions, set out

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in Section VII so neither is credited with the other's.

**Problem.** Given a circuit DAG on $n$ logical qubits, a machine of $K$ cores with intra-core coupling graphs, inter-core links and designated communication ports, and an initial layout, emit a legal operation stream realising every gate while minimising the number of EPR pairs consumed. Section II-B states the cost model this objective assumes and what it excludes.

**Contributions.** We make three.

(i) *The* PORTER *compiler*: a plan-oriented remote stage that scores a complete inter-core action before any preparation SWAP for it exists and commits before preparing (Section III-D), a capacity discipline that treats free slots as a legality condition and manufactures them by a two-tier relief search (Section III-G), and a deterministic initial layout replacing the best-of-$N$ search its predecessors run (Section III-A).

(ii) *Per-circuit selection of the lookahead horizon.* Every SABRE-descended router fixes this parameter by constant, and no tested constant is best across our benchmark — the best value spans a factor of thirty-two. PORTER instead measures how dense the circuit's interaction graph is, uses that to prune a three-value candidate set, routes each and keeps the lowest-EPR result. The configured horizon always competes and wins ties, so the selection cannot return a worse route than routing once. It is what carries PORTER past DSABRE on the scale-up series (Section IV-F1).

(iii) *A controlled study over forty-eight instances*, under native, shared-layout and cross-layout protocols, with ablations that measure what each mechanism contributes — including two negative findings, which Section VI interprets. Every route is certified by a separately implemented replay verifier, and the harness reproducing every table is released.

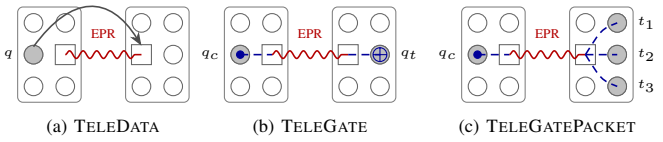
Section IV-A states the four questions the evaluation answers. Each mechanism is tied to the experiment that assesses it: plan commitment to the hop-wise ablation, capacity relief to the completion ablation, the initial layout to the shared- and cross-layout controls, horizon selection to the scale-up series and the star-Multiplier livelock, packets to the action-vocabulary study, and the verifier to fault injection.

The PORTER implementation, the replay verifier, the full cross-router comparison harness, and the raw per-circuit results are publicly available at https://github.com/ebony72/porter.

## II. Model, Primitives, and Problem Formulation

### A. Quantum Circuits and Routing

A circuit acts on $n$ logical qubits. We work in the basis $\{\texttt{id}, \texttt{rz}, \texttt{sx}, \texttt{x}, \texttt{cx}\}$ of MQT-Bench's Qiskit-opt3-transpiled circuits [9]: arbitrary one-qubit gates plus the two-qubit CX. A circuit is a directed acyclic graph, with a dependency edge between consecutive gates on each logical qubit. The *front* is the set of gates with no unexecuted predecessor; only front gates can execute or be routed toward execution.




(a) TeleData (b) TeleGate (c) TeleGatePacket

Fig. 1. The three communication primitives; each consumes one EPR pair between the two communication ports (squares). (a) TeleData moves qubit $q$ (which must sit adjacent to its free port) onto the free destination port. (b) TeleGate executes one remote CX through a cat-entangler [7] (Appendix E): the target-side port acts as a proxy for the control $q_c$; neither qubit moves. (c) TeleGatePacket brackets an ordered run of remote CX gates sharing the control $q_c$ between one cat-entangler/disentangler pair, amortising a single EPR pair over the whole run; each target is routed next to the port in turn while the ports stay reserved.

Physical qubits are partitioned into *cores*. Each core has an intra-core coupling graph, and pairs of cores may be joined by an inter-core link whose endpoints are *communication ports*. A *layout* maps every logical qubit to a physical position. A two-qubit gate is *local* if both operands share a core and *remote* otherwise: a local gate that is not yet adjacent needs intra-core SWAPs, a remote gate needs an inter-core primitive. We write $\kappa_\ell(q)$ for the core holding $q$ under layout $\ell$, $d_{\mathrm{L}}(c, p, p')$ for intra-core distance and $d_{\mathrm{C}}(c, c')$ for core-graph distance.

### B. Objective, Scope, and Assumptions

**Objective.** Given a circuit DAG $C$ on $n$ logical qubits, a machine $\mathcal{A}$ of $K$ cores with intra-core coupling graphs, inter-core links and designated communication ports, and an initial layout $\ell_0$, produce a legal operation stream that realises every gate of $C$ while minimising the number of EPR pairs consumed. Intra-core SWAP count, circuit depth and compile time are recorded but are not part of the objective and are never traded against EPR pairs by any rule in this paper.

**Cost model.** One EPR pair is consumed per TeleData, per TeleGate, and per TeleGatePacket — one per packet, however many remote CX gates the packet amortises — and a pair is charged the same regardless of which link carries it. A multi-hop relocation is a sequence of single-hop TeleData actions and is charged once per hop. This is the model TeleSABRE [10] and DSABRE [11] use, which is what makes the three compilers comparable in a single unit.

Communication ports are ordinary physical qubits, occupied for the duration of an action and free otherwise; no core is assumed to have dedicated communication memory. What the model leaves out — link contention, entanglement latency and fidelity, purification, classical communication cost — and what that means for the conclusions drawn here is set out in Section VI-A.

### C. Communication Primitives

Three inter-core primitives are in play, sketched in Figure 1; each consumes exactly one EPR pair.

**State teleportation (TeleData).** Consumes one EPR pair to move a logical qubit from a communication port in one core to a free port in another, after which every later gate that qubit meets in its new core is local. Its value is therefore *latent*: it

is paid once and repaid over however many future gates the relocation makes local.

**Gate teleportation (TELEGATE).** Executes a single remote CX with neither operand moving. One EPR pair entangles the two communication ports; a *cat-entangler* — $\mathrm{CX}(q_c, e_A)$, measurement of $e_A$, and a classically controlled $X$ on $e_B$ — leaves $(q_c, e_B)$ in the cat state $\alpha|00\rangle + \beta|11\rangle$, so every CX controlled by the remote port $e_B$ acts as if controlled by $q_c$. A *cat-disentangler* — Hadamard on $e_B$, measurement, a classically controlled $Z$ on $q_c$ — restores $q_c$ and frees both ports (Appendix E). Its value is entirely *direct*: exactly one gate, and nothing is left behind.

**Same-control packets (TELEGATEPACKET).** When several remote CX gates share the same control and their targets lie in one core, the run can be bracketed between a *single* cat-entangler and disentangler: the ports stay reserved, each target is routed beside $e_B$ in turn, and the whole run costs one EPR pair instead of one each. This is the amortisation idea of AutoComm [8], specialised to the shared-control case and applied inside a routing loop rather than as a pre-pass.

The three differ in one respect that matters for scoring: how many gates one pair is spread over. TELEDATA amortises over an unknown future, TELEGATE over exactly one gate, TELEGATEPACKET over a run whose length the compiler must discover. Section III-E makes that a direct/latent/lookahead decomposition.

## III. THE PORTER COMPILER

Given a circuit and an initial layout $\ell_0$ (which PORTER builds itself, Section III-A), PORTER repeats the following loop (Algorithm 1) until every gate has been emitted or a configurable iteration budget is exhausted. Algorithm 1 charts one iteration; its shaded region is the plan-first remote stage that distinguishes PORTER from compilers deciding at single-action granularity; Appendix F walks through it on a concrete front where it changes the outcome.

**Algorithm 1** PORTER `compile`$(C, \ell_0)$

$\ell \leftarrow \ell_0$; initialise DAG active/frontier sets from $C$
**while** DAG has active gates **do**
  $F \leftarrow$ DAG front (bounded by `dag_front_limit`)
  classify $F$ into singles, direct-local CX, local-CX-needing-SWAP, adjacent-remote CX, non-adjacent-remote CX *(Sec. III-B)*
  **if** singles or direct-local CX present **then**
    execute them all; remove from DAG; **continue**
  **end if**
  **if** local-CX-needing-SWAP present **then**
    generate candidate intra-core SWAPs, apply the best-scoring one (Eq. 1); **continue** *(Sec. III-C)*
  **end if**
  generate normal remote plans $\mathcal{P}$ and blocked-teleport plans $\mathcal{B}$ *(Sec. III-E)*
  **if** $\mathcal{P} \neq \emptyset$ **then**
    $p^* \leftarrow$ best ready-or-preparable plan in $\mathcal{P}$
    **if** some $b \in \mathcal{B}$ has gain $\gg$ gain$(p^*)$ **then**
      $p^* \leftarrow$ best opportunistic capacity-relief plan for $b$ *(Sec. III-G)*
    **end if**
  **else**
    BFS from each blocked target core to its nearest reservoir; $p^* \leftarrow$ best tail-relief plan *(Sec. III-G)*
  **end if**
  **if** $p^*$ is `None` **then**
    **report deadlock; abort**
  **end if**
  apply $p^*$'s own preparation SWAPs until ready, then execute it
**end while**
**return** emitted operations, final layout, stats

### A. Initial Layout

PORTER routes from any initial layout, and the shared- and cross-layout experiments of Section V-B run it on layouts built by other compilers. Left to itself it builds $\ell_0$ with one deterministic heuristic — no external partitioner, no best-of-$N$ search — giving every core at least $f_{\min}$=2 free positions.

That floor is a natural requirement rather than a tuned parameter, and DSABRE [11] independently arrives at the same value as its own routing legality condition. A core with no free physical qubit is dead: it cannot vacate an occupied communication port by local SWAPs, so no inter-core action can reach it. A core with exactly one free position is alive but cannot accept a TELEDATA, since the arriving qubit would fill it. Two is therefore the smallest reserve under which every core can both receive a teleport and remain able to make room afterwards. $f_{\min}$ accordingly plays three roles, which we keep apart: this *initial* condition; the *admission threshold* a destination must meet *before* a normal TELEDATA (Section III-D), so at least one position remains after one; and the threshold defining a *reservoir* for capacity relief (Section III-G). It is not maintained after every move: a core admitted at exactly $f_{\min}$ drops to one free, and cores do reach one free position during routing.

**Temporal locality-greedy.** Three steps. *(i) Weighting*: each CX gate $g = (q_i, q_j)$ contributes $\gamma^{\mathrm{asap}(g)}$ to the interaction weight $w(q_i, q_j)$, where $\mathrm{asap}(g)$ is $g$'s as-soon-as-possible layer in the CX-only DAG and $\gamma$=0.99, so early interactions weigh more than late ones while repeated interactions accumulate. *(ii) Core assignment*: logical qubits are visited in the order they first appear as a CX operand and each is placed in the core holding its heaviest already-placed neighbour when that core has headroom above $f_{\min}$, falling back to the core with the most remaining reserved capacity. *(iii) Physical placement*: within each core, logicals are laid onto positions by descending interaction weight onto ascending intra-core centrality, so the most-connected qubits take the most central slots.

The steps are complementary. The discount co-locates the pairs that must interact soonest; first-appearance order handles chain-structured circuits, where a balanced-first-layer rule would scatter the chain across cores; and the centrality step keeps hub qubits cheap to reach. Its one weakness is that the

$f_{\min}$ reservation is uniform across cores, which suits a grid or a ring but not a core graph with a hub (Section V-E).

### B. Front Classification

Every front gate is classified under the current layout $\ell$. 1Q gates are always directly executable; a CX $g = (q_i, q_j)$ with both operands in core $c$ is *direct-local* if $d_{\mathrm{L}}(c, \ell(q_i), \ell(q_j)) = 1$ and *local-needing-SWAP* otherwise; a CX whose operands lie in different cores is *adjacent-remote* if those cores are core-graph neighbours and *non-adjacent-remote* otherwise. Directly executable gates drain first and completely, intra-core SWAPs are exhausted next, and remote-stage decisions are taken only once no front gate is intra-core. A cost guard caps the front at 96 gates; it binds only on the three largest QPEexact instances, where removing it leaves the emitted stream unchanged, since truncation is by ascending gate index and directly executable gates drain before any routing decision is taken.

### C. Intra-Core SWAP Scoring

This stage is SABRE [12]. For a local-needing-SWAP gate set $F_{\mathrm{loc}}$, candidate SWAP edges are the coupling-graph neighbours of every physical position $F_{\mathrm{loc}}$ touches, and each is scored under its post-SWAP layout $\ell_a$ by

$$J(a) = \bar{d}_{\mathcal{F}}(a) + \lambda_{\mathrm{loc}}\, \bar{d}_{\mathcal{E}}(a), \qquad \lambda_{\mathrm{loc}} = 0.8, \tag{1}$$

the mean intra-core distance $d_{\mathrm{L}}(c, \ell_a(q_i), \ell_a(q_j))$ over $F_{\mathrm{loc}}$, plus the same mean over an extended set of up to $N_{\mathrm{loc}}$=20 local CX gates from the next $L_{\mathrm{loc}}$=12 DAG layers. The lowest $J$ is applied; ties break on raw front-layer distance, then on a constant-seeded generator. Note the sign convention differs between the two stages: local candidates *minimise* distance, whereas remote plans *maximise* gain (Section III-E).

### D. Plan-First Remote Stage

PORTER manipulates remote work as *plans*. A plan is a complete, committable specification of one inter-core action: which primitive fires, which gate or same-control run it resolves, its direction, and consequently its *readiness conditions* — which logical endpoint must sit adjacent to which communication port, and which ports must be free. What it deliberately leaves unspecified is the SWAP sequence that establishes those conditions. Alternatives are compared at the level of *what* remote work to do next, each scored as if its preconditions were met; only the committed plan is handed to the machinery that works out *how*.

Formally, a plan is a tuple

$$p \;=\; \big(\pi,\; G(p),\; \mathrm{dir},\; c_{\mathrm{src}},\; c_{\mathrm{dst}},\; \Pi(p),\; \mathrm{ready}(p, \ell)\big),$$

with $\pi \in \{\textsc{TeleData}, \textsc{TeleGate}, \textsc{TeleGatePacket}\}$ the primitive, dir its orientation, $c_{\mathrm{src}}$ and $c_{\mathrm{dst}}$ the cores involved, $\Pi(p)$ the communication ports it reserves, and $\mathrm{ready}(p, \ell)$ the predicate that holds when $p$ can fire. $G(p)$ is the set of gates the action *discharges*: for a TeleGate, the single front CX it executes; for a TeleGatePacket, the run of same-control gates collected in Section III-F; for a TeleData, the empty set, since moving a qubit executes no gate — its value lies entirely in the gates the relocation makes local, which the scorer estimates (Section III-E).

Three quantities are kept distinct: the plan's *utility*, evaluated under the hypothesis that $\mathrm{ready}(p, \ell)$ holds whether or not it does; its *readiness cost*, the local work still needed, used only to break ties; and the *preparation policy* that chooses those local SWAPs after $p$ is committed. Two design decisions follow: utility never includes readiness cost, and once a plan is committed PORTER works only on that plan — it generates no other plan and scores no alternative — until the plan fires or is abandoned because no preparation SWAP improves its readiness. Section VII places both against the baselines' choices; Section V-C measures the second.

**The candidate sets.** When the front holds only remote CX gates, PORTER builds two sets. **Normal plans** $\mathcal{P}$ are the actions that could fire once prepared. For each adjacent-remote CX they hold a TeleGate; a TeleGatePacket if a same-control run of $\geq 2$ gates can be collected; and, for either endpoint, a TeleData into the partner's core when that core has $f_{\min}$ free positions. A non-adjacent-remote CX cannot be discharged in one action, so it contributes only TeleData hops that move one endpoint to a neighbouring core *strictly closer* to the other endpoint's core, again subject to $f_{\min}$.

**Blocked plans** $\mathcal{B}$ are the same TeleData candidates whose destination fails the $f_{\min}$ test. PORTER never commits one — executing it would drive a core below the floor. It scores them anyway, as a measurement of what capacity is currently costing: $\mathcal{B}$ names the cores that are saturated and says how much gain is being forgone there, which is what the relief tiers of Section III-G act on.

**Selection.** Every plan is scored (Section III-E) *before* any preparation SWAP for it exists. PORTER sorts $\mathcal{P}$ by the key below and takes the best plan that is either ready or admits an improving preparation SWAP. It then overrides that choice in two cases, both driven by $\mathcal{B}$: if the best blocked plan outscores it by more than $m_{\mathrm{assist}}$=5.0, PORTER instead relieves that saturated core; and if $\mathcal{P}$ is empty, it relieves the blocked destinations in ascending hop order. In both cases what is committed is a relief TeleData generated for the purpose, not the blocked plan itself. Having committed, PORTER applies only that plan's preparation SWAPs until it is ready, no improving SWAP remains, or the decision state — plan, layout, RNG — repeats, which raises rather than loops.

**Preparation SWAPs.** A committed plan has exactly two readiness requirements, one per side: source and destination for TeleData, control and target for TeleGate and TeleGatePacket, whose target side is routed at execution time (Section III-F). For each unmet requirement, candidate SWAPs are the edges incident to the plan's logical endpoint, or to a nearest free position tied for minimum distance to the relevant port while that port stays occupied. Each is scored by the post-SWAP readiness cost, and at most one improving SWAP per side is applied per pass.

### E. Plan Scoring: Direct, Latent, and Lookahead Gain

Every plan $p$ scores $\mathrm{total}(p) = \mathrm{direct}(p) + \mathrm{lookahead}(p)$. All kinds share one convention. Write $\kappa$ for the map sending

a logical qubit to the core holding it under the current layout, and $\kappa'$ for the same map under the layout the plan would produce — identical to $\kappa$ except that a TELEDATA's moving qubit now sits in $c_{\mathrm{dst}}$. For a CX $g = (q_i, q_j)$,

$$\Delta(g) = d_{\mathrm{C}}\big(\kappa(q_i), \kappa(q_j)\big) - d_{\mathrm{C}}\big(\kappa'(q_i), \kappa'(q_j)\big)$$

is the core-graph distance the plan removes from $g$: 1 for a gate the move makes local, 0 for one it does not touch, negative for one it pushes apart.

A **TELEGATE** scores a constant direct gain of 1, one remote CX resolved, with no lookahead. A **TELEGATEPACKET** scores the number of remote CX gates in the collected packet. These are not discounted lookahead: collection returns a committed executable sequence, and once the cat resource is live every member executes under it, each target reached by local SWAPs that are forbidden from touching the two reserved ports (Section III-F). A **TELEDATA** plan executes no gate, so its whole score is an estimate of the gates its moving qubit $q$ will meet. Three sets of CX gates on $q$ contribute, found by simulating the DAG frontier forward under $\kappa'$:

- $D(p)$, the front CX on $q$. Because $q$ has at most one zero-predecessor gate at a time, this is the plan's seed gate and nothing else, so $|D(p)| = 1$.
- $L(p)$, the CX gates on $q$ exposed by *local closure*: the simulator repeatedly retires every front gate that is executable under $\kappa'$ — one-qubit gates, and CX gates whose operands $\kappa'$ places in one core — and every CX on $q$ that surfaces while it can do so joins $L(p)$. Closure stops at the first simulated front with nothing locally executable. These are current pressure rather than prediction: no gate outside $\kappa'$'s reach was needed to expose them, and the teleport fixes their core distance the moment they appear.
- $A_k(p)$, the CX gates on $q$ found by continuing past that stopping front, in simulated layer $k \le L_{\mathrm{tp}}$, discounted by $w_k = \max(w_{\min}, w_0\gamma^{k-1})$ with $w_0$=0.9, $\gamma$=0.9, $w_{\min}$=0.5. These are prediction.

Then

$$\begin{aligned} \mathrm{direct}(p) &= \sum_{g \in D(p) \cup L(p)} \Delta(g), \\ \mathrm{lookahead}(p) &= \sum_{k \ge 1} \sum_{g \in A_k(p)} w_k\, \Delta(g), \end{aligned} \tag{2}$$

so $L(p)$ enters at full weight alongside $D(p)$. Alongside the score PORTER also computes $\mathrm{cap}(p) = 1/\max(1, f_{\mathrm{dst}} - 1)$, which grows as the destination approaches the capacity floor. It is never added to total; it is carried separately and used only to break ties, so that between two plans of equal estimated value the one leaving more room wins.

Preparation cost appears nowhere in total. Plans are ordered by the lexicographic key

$$\big(-\mathrm{total},\ \neg\mathrm{ready},\ \mathrm{prep},\ \mathrm{rank}(\pi),\ \mathrm{cap},\ -\mathrm{lookahead}\big),$$

smallest first, so gain decides first, readiness second, and preparation cost third; a stable plan identifier resolves anything still tied. The kind rank prefers TELEDATA (0) to a packet (1) to a single TELEGATE (2), keeping the layout-changing action when estimated value, readiness and preparation cost are all equal. Because preparation cost sits below gain, a plan that is cheap to prepare cannot displace one that buys more progress — which is the property that distinguishes this scorer from DSABRE's, where staging cost is additive (Section VII). The horizon $L_{\mathrm{tp}}$ counts *layers*, and at most one gate on the moving qubit can appear per layer, since a qubit has at most one zero-predecessor gate at a time. A gate cap $N_{\mathrm{tp}}$ therefore cannot bind whenever $N_{\mathrm{tp}} \ge L_{\mathrm{tp}}$, so $L_{\mathrm{tp}}$ alone governs how far the estimate looks. It is the one parameter PORTER does not fix by constant; choosing it is a protocol question, deferred to Section IV-F1.

### F. Same-Control Packet Collection

Same-control CX gates are chained by DAG dependency edges, so they never appear in the front together: a packet cannot be read off the front and must be collected by forward simulation. From a seed adjacent-remote CX $g = (q_c, q_t)$ with target core $c_t$, PORTER simulates the DAG frontier forward for up to 120 layers, absorbing any front CX that shares control $q_c$ and has its target in $c_t$ (up to 8 gates), and any front 1Q gate whose qubit is already in $c_t$, since those execute freely while the cat resource is live. Collection stops at the first unproductive layer or at the size cap. The window counts simulated layers, not collected CXs, so it acts as a cost guard rather than a tuning knob. Because the simulation reaches past the current front, a packet may contain gates a front-limited scan could not. Collection is a heuristic in one respect we have not ablated: stopping at the first unproductive layer can miss a profitable packet separated by harmless intervening work.

Collection only *discovers* the run; nothing is emitted while it simulates. The sequence it returns is recorded on the plan, and if that plan is committed the sequence is emitted in exactly that order — one `cat_entangler`, which consumes the packet's single EPR pair; one `cat_cx` per member, each target first brought beside the port by ordinary local SWAPs that may not touch either live port; and one closing `cat_disentangler`, which frees both ports. Fixing the order at collection time is what lets the scorer count every member as work the packet will actually do: the members are DAG-chained, so executing them in that order respects their dependencies, and no member can become unexecutable once the resource is live.

### G. Capacity Relief

Both tiers answer the same question — some core $c$ is at or below $f_{\min}$ and a TELEDATA wants to enter it — by moving a qubit *out* of the congested region rather than into it. Neither picks a destination freely: a breadth-first search from $c$ over the core graph finds the nearest *reservoir*, a core with at least $f_{\min}$ free positions, and the relief action moves one qubit from the penultimate core of that path into the reservoir at the end of it. Every logical qubit in the penultimate core is a candidate, and the candidates are scored and ordered by the plan machinery already described, grouped by hop count so that shorter paths are tried first. Chaining beyond one hop is optional and enabled by default, which lets relief propagate along a path when the immediate neighbour is itself full.

The two tiers differ only in when they fire. **Tail relief** fires reactively, when $\mathcal{P}$ is empty and the front is starved. **Opportunistic relief** fires pre-emptively, while a normal plan is still available, whenever a blocked plan out-scores it by more than $m_{\text{assist}}$ — the judgement being that clearing the congestion now is worth more than taking the best currently legal action.

### H. Progress and the Failure Model

Deterministic greedy routing on a capacity-constrained machine can stall. PORTER layers four defences against that.

*(i) Keep capacity from reaching zero.* A core with no free physical qubit is terminal: it cannot vacate an occupied communication port by local SWAPs, so no inter-core action can reach it. PORTER legalises the entry layout so no core starts full, and admits a TELEDATA into core $c$ as a normal plan only while $\text{free}(c) \geq f_{\min}$, leaving at least one free position behind. Routing therefore cannot drive a core into the terminal state; only an external layout that already fills one can present it, and that PORTER reports as infeasible rather than routing.

*(ii) Detect starvation rather than thrash.* An empty $\mathcal{P}$ alongside a non-empty $\mathcal{B}$ signals that the front is starved by capacity, and names the cores responsible. *(iii) Manufacture capacity.* Tail relief turns that signal into work (Section III-G). *(iv) Bound every loop.* Plan sides with infinite preparation cost are never selected; the compiler hashes its decision state once per preparation round and raises if it repeats; and two budgets bound the loops from outside.

These defences give two of the three guarantees a router might offer. *Correctness*: a returned operation stream realises every gate and violates no coupling or port rule. *Termination*: every run ends, in success or by raising a diagnostic or tripping a guard. *Completion*: that a run ends in success. PORTER provides the first two by construction. It does *not* provide the third, and has no analogue of DSABRE's termination-with-success proof under stated architectural hypotheses (Section VII). Its completion record is empirical, and Section V reports it.

A per-iteration cost analysis of the loop is given in Appendix I.

## IV. EXPERIMENTAL METHODOLOGY

### A. Research Questions

The evaluation answers four questions, each with its own controls, and Section V is organised around them.

**RQ1** **End-to-end cost.** Under each compiler's own recommended protocol, how does PORTER's EPR consumption and compile time compare with TELESABRE's and DSABRE's?

**RQ2** **Routing quality with the layout controlled.** Given the *same* externally supplied initial layout, how does PORTER's routing loop compare with DSABRE's — first at an equal route count, then when each is allowed its own refinement schedule?

**RQ3** **Mechanism attribution.** What does each of PORTER's mechanisms — plan commitment, capacity relief, the packet primitive, the TELEGATE family, the initial layout and horizon selection — contribute on its own?

**RQ4** **Robustness.** How do the answers change with circuit width, core count and core-graph shape?

Each results subsection opens with the question it answers, what is held fixed, and the metric; the interpretation of the answers, including the places where they appear to conflict, is collected in Section VI.

### B. Terminology

Budget comparisons depend on using these terms consistently, so we fix them here and use no synonyms.

- A **route** (equivalently a *routing pass*) is one invocation of the routing loop of Section III on one circuit from one layout, producing one operation stream.
- A **candidate layout** is one initial mapping a protocol supplies to the router. A protocol's cost is its layout count times its routes per layout.
- A **protocol** is a named row of Table I: a layout source, a route budget and a selection rule. We avoid "attempt", which the baselines use for both meanings.
- A **candidate horizon** is one value of $L_{\text{tp}}$ offered to horizon selection; selecting among them costs one route each.
- **Complete compile cost** is the wall-clock time of an entire protocol — layout generation plus every route over every candidate — not of the winning route alone.

For the machine and the router we use: *core*, *communication port*, and *reservoir* (a core holding at least $f_{\min}$ free positions); a front CX is *local*, *adjacent-remote* or *non-adjacent-remote*; a plan is *normal* or *blocked*, and *ready*, *preparable* or neither. A run that does not complete does so in one of four ways, distinguished throughout: a capacity *deadlock* leaves no legal plan and no relief path, and raises a diagnostic; a *livelock* makes local progress indefinitely while retired gates and EPR count stand still, and is stopped by the iteration guard; a *timeout* exceeds the wall-clock cap; and an *infeasible input* is an external layout that already fills a core, reported rather than routed (Section III-H).

### C. Implementation and Verification

PORTER is implemented in Python and released with the verifier and comparison harness. It takes no seed: heuristic ties resolve through a fixed chain of deterministic keys, any residual tie through a constant-initialised generator, and every residual tie-break is counted in the emitted statistics. In the released software environment, one route on a fixed circuit, hardware and revision therefore emits a byte-identical stream across repeated runs, and we certify behaviour preservation across refactorings by hashing it; we have not tested identity across Python versions or platforms. One caveat applies to horizon selection (Section IV-F1): its speculative routes share a wall-clock budget, so on a loaded machine a candidate may be abandoned that would otherwise have won. Each route is exactly reproducible; the protocol is reproducible only up to that budget, and the horizon a run selected is recorded with its result. Compile time is kept low by incremental data structures

— chiefly a copy-on-write DAG frontier simulator whose snapshots cost $O(|\text{frontier}|)$ rather than $O(|\text{remaining circuit}|)$ — together with cached distance tables, $O(1)$ free-slot accounting, and apply/restore SWAP scoring in place of layout cloning.

Every reported PORTER run is certified by a replay verifier written separately from the router. It shares no routing, scoring or layout code and calls nothing in the routing loop: from the emitted operation stream, the original circuit and the initial layout alone, it re-derives gate coverage, per-qubit dependency order, coupling-edge legality of every local operation, port rules for every teleport, telegate and packet, and the consistency of every reported statistic. The verifier is itself tested by fault injection — ten classes of mutation applied to passing streams, every mutant rejected (Appendix D). Baseline routes are *not* verified this way: TELESABRE and DSABRE emit their own operation formats, and we report the EPR counts their harnesses produce, as their published evaluations do.

### D. Setup

**Architectures.** We use the two grid-of-grids topologies DSABRE [11] evaluates on: architecture A, a $2\times2$ array of $3\times3$ cores (36 physical qubits, four links), and architecture H, a $2\times3$ array of $4\times4$ cores (96 physical, seven links). Sections V-D and V-E add its $5\times5$-core scalability series and two networks of IBM 27-qubit heavy-hex tiles. All are read from the baselines' own device files, so every compiler sees identical physical qubit ids.

**Benchmarks.** We use the same MQT-Bench [9] circuit files DSABRE's evaluation uses, confirmed by exact CX-count match: a 25-qubit suite (AE, GHZ, Graphstate, QFT, QNN, Random) on architecture A, and a 64-qubit suite adding QPEexact, QAOA and Multiplier on architecture H, with QNN the deeper 8,126-CX variant. We add an 80-qubit suite of the same nine families. Its AE and QNN instances are not produced by the stock generator and were reconstructed to match its construction: QNN is the 12,718-CX ZZ-feature-map circuit rather than the 79-CX chain the current generator emits, whose interaction graph matches GHZ's. Appendix A gives both reconstructions, their validation and the CX counts; the scripts and files are released.

**Baselines.** We compare against TELESABRE [10] and DSABRE [11], the latter in its default `dSABRE_BurstExt` mode with its shipped configuration, which is *safe mode*: capacity as a legality condition with a per-core reserve, BFS slack relay, and a guaranteed per-gate recovery transaction (Section VII). Every DSABRE number here was generated from revision `cb5515b` (2026-08-21), and the harness that produced them is released with PORTER.

Three protocol decisions avoid disadvantaging DSABRE. First, we run it under its own budget convention, letting its three recovery budgets derive from the architecture as its own drivers do; a binding iteration budget would cut its heuristic search short and hand the remainder to its guaranteed drain at a worse EPR count. Second, we do not charge it for pre-routing layout repair. In safe mode it brings an external layout up to its reserve before the clock starts, and those relay teleports land in its EPR counter although they adjust the starting layout rather than route anything; PORTER performs the same repair as a free permutation, so we net it out for both. Third, we hand it both families of its layout candidates rather than picking one (Section IV-E).

TELESABRE's stock C binary crashes deterministically on architecture A, in its Hungarian initial-layout pass rather than in routing: a hard-coded capacity threshold leaves some qubits unplaced and their sentinel index is then dereferenced. All TELESABRE numbers use a minimally patched binary that falls back to the least-loaded core. The patch changes placement only for qubits the stock binary leaves unplaced, and architecture-H results are unchanged by it; it is in our repository. Every compiler invocation is capped at 900 s wall-clock, and a call exceeding it is recorded as a failure to complete.

### E. Initial-Layout Protocols

We evaluate under three initial-layout protocols, designed to separate routing quality from initial-layout quality.

**Own-protocol comparison.** Each router is first evaluated under its own established best-practice protocol. For the two baselines that is best-of-three, matching the convention already used to compare TELESABRE and DSABRE to each other [11]: TELESABRE runs its built-in Hungarian initial-layout heuristic at config seeds 0–2, and DSABRE runs Qiskit `SabreLayout` on its corners-removed architecture graph at seeds 0–2, each layout routed by SABRE-style forward → backward → forward passes; both report the best EPR count. PORTER instead builds one deterministic temporal locality-greedy layout (Section III-A) and routes it at each of the three candidate lookahead horizons of Section IV-F1, keeping the best — three routing passes over one layout, against nine to eighteen over three to six. It searches no layouts, so every number is reproducible from the circuit file alone.

DSABRE's corner removal reserves low-degree physical qubits as communication slots by excluding them from the SabreLayout coupling map. It ships two variants of the rule and neither dominates on our architectures, so rather than choose we give DSABRE both families and take its best: up to six candidate layouts instead of three. This can only strengthen the baseline relative to its own published protocol, and it is why our DSABRE numbers sit at or below the figures it publishes for the architectures it reports.

**Shared-layout comparison.** To separate layout quality from routing quality, both DSABRE and PORTER are then run from an identical externally supplied mapping — TELESABRE's own initial layout, and separately `pytket-dqc`'s hypergraph partition (Appendix B) — so the only difference left is the routing loop. **Cross-layout comparison.** Finally each compiler routes the other's own layout, which closes the remaining cell of the design.

### F. Operating Protocols

Section III defines one routing pass. Three protocols run that pass more than once and keep the best result by EPR; because

they invoke the router repeatedly rather than changing what a single pass does, we describe them here rather than as part of the algorithm. All three are deterministic given the circuit, the hardware and the configuration, and none takes a seed.

*1) Lookahead-horizon selection:* The teleport-lookahead horizon $L_{\mathrm{tp}}$ is the one parameter of Section III-E for which no tested constant is best across the corpus. In an offline sweep over $\{4, 8, \ldots, 1024\}$ on every instance the best value spans the whole range: 80-qubit QNN wants 4, the 25-qubit suite wants 16, and every 100-, 200- and 360-qubit instance wants 128, beyond which EPR saturates. The gap between the shipped 16 and the per-instance best is 17.2% geometric-mean EPR.

Interaction density sets the direction. Write $\rho$ for the mean fraction of the other logical qubits each qubit shares a CX with. As $\rho \to 1$ a long horizon reports the moving qubit as needed in every core at once, so the gain estimate degenerates into noise and short horizons win; when $\rho$ is small the signal is coherent and long horizons win. Across the benchmark $\rho$ ranks the best horizon at Spearman $-0.765$, but it is not a usable predictor alone: in the band $0.42 \le \rho \le 0.53$, where most 64-qubit circuits sit, QAOA wants 8 and AE wants 128, and a threshold rule on $\rho$ recovers only 8.0% of the available 17.2%.

PORTER therefore uses $\rho$ to *prune* rather than to choose. A circuit with $\rho \ge 0.80$ is routed at $L_{\mathrm{tp}} \in \{4, 8, 16\}$, every other circuit at $\{8, 16, 128\}$, and the lowest-EPR route is kept. No instance above the threshold ever selected the long horizon, so pruning costs no EPR and removes the dominant compile-time term: a long horizon that fails to converge and consumes the whole protocol's budget. Two rules bound the cost. The circuit's configured horizon is routed first, unguarded, and wins ties. The remaining candidates share a wall-clock budget proportional to what that first route cost, and one that overruns it is abandoned. The guarantee this gives is conditional and worth stating exactly: *provided the configured-horizon route completes*, selection returns no higher an EPR count than that route would alone. It can still raise compile time, and which speculative candidates finish — hence which route is selected — depends on the wall-clock budget and so on the machine (Section IV-C).

**This rule was fitted on the corpus it is evaluated on.** The sweep, both candidate sets and the 0.80 threshold were fixed by inspecting the 48 benchmark instances, and there is no held-out family, so evidence for the *size* of the gain is in-sample. Two things bound the risk. The rule is a portfolio, not a prediction: the configured horizon always competes and wins ties, so a poor candidate set costs compile time but cannot worsen the selected route. And the direction the sweep exposes — that no tested constant is best across circuits whose densities differ by an order of magnitude — does not depend on the values chosen. An out-of-sample validation remains to be done.

*2) Backward refinement (*PORTER+*R):* DSABRE routes each candidate layout with a forward → backward → forward schedule, so a protocol that matches its routing effort rather than only its layout count is needed to separate the two effects. PORTER+R supplies one. The schedule is SABRE's bidirectional layout refinement [12] applied to whole routes. Let $R(C, \ell)$ denote routing $C$ from $\ell$ and $\ell'$ the layout it ends on. Pass 1 computes $R(C, \ell_0)$; pass 2 computes $R(C^{\mathrm{rev}}, \ell'_1)$ on the reversed circuit and *discards its route*, keeping only $\ell'_2$; pass 3 computes $R(C, \ell'_2)$. Reversing $C$ reverses every dependency edge, so routing it from the forward pass's end state yields a layout adapted to the *start* of the forward circuit. Odd pass counts above three repeat the cycle; $P$=5 throughout. Because the best forward pass is selected, PORTER+R can never report a higher EPR count than the pass 1 route it contains — at the cost of $P$ routes where one would do. It adds no randomness.

One property of the schedule matters for reading its results. Refinement consumes a post-routing layout, and routing destroys the per-core free-slot reservation $\ell_0$ was built with (Section III-A), so an intermediate pass may start from a layout worse than $\ell_0$ for PORTER's own capacity machinery. Individual passes therefore regress on some instances, and only the best-pass rule keeps that from surfacing.

*3) Layout-discount search (*PORTER+*S):* PORTER+S searches the one free scalar in the layout heuristic — the per-layer discount $\gamma$ of Section III-A, fixed at 0.99 elsewhere — in two phases.

*Phase 1* builds a layout for each $\gamma \in \{0.80, 0.90, 0.95, 0.99, 1.0\}$ and routes it once at the configured horizon, without selection: five routes. The lowest-EPR layout wins. *Phase 2* runs the $P$=5 refinement schedule above from two seed layouts — phase 1's winner, and the $\gamma$=0.99 layout when that is not already the winner — adding $P-1$=4 routes each. Refining the winner is not reliably best: on 64q QNN, $\gamma$=1.0 wins phase 1 at 604 and refines no further, while refining the $\gamma$=0.99 layout reaches 537, so both are run.

The total is therefore $5 + 4 = 9$ routes when $\gamma$=0.99 wins phase 1 and $5 + 4 + 4 = 13$ otherwise, which is the 9–13 of Table I; a schedule that converges early can stop between the two. $\gamma$=0.99 is always a candidate, so PORTER+S cannot return a higher EPR count than the configured single route.

PORTER+SR runs horizon selection and the $\gamma$-search and keeps the lower EPR of the two, netting the one route they share. The two look for different things — the horizon at a fixed layout against the layout at a fixed horizon — so neither subsumes the other.

### G. Comparison Protocols and Budgets

Table I states, for every configuration that appears in the results, where its initial layouts come from, how many routing passes it spends, and what it selects on. We use the names in its first column throughout.

Three notions of a matched budget are in play and none is universally fair, so we report against more than one. Equal *route counts* (shared-layout PORTER against shared-layout DSABRE, three each) equalise how often the router runs. Equal *layout budgets* put PORTER on DSABRE's candidate set. Equal *wall-clock* is not enforced anywhere; instead every table reports complete compile time so a reader can renormalise. PORTER costs more per route than DSABRE (Section V-F), so equal route counts favour it on time and not on effort, and equal layout budgets do the reverse.

TABLE I

EVERY CONFIGURATION THAT APPEARS IN SECTION V: LAYOUT SOURCE, LAYOUT COUNT $|\mathcal{L}|$, ROUTING PASSES PER LAYOUT $P$, TOTAL ROUTES $|\mathcal{L}|\cdot P$, AND THE QUANTITY MINIMISED OVER THEM. LAYOUT COUNTS MARKED [a] ARE 6 AT 25 AND 80 QUBITS AND 3 AT 64, WHERE DSABRE'S TWO CORNER RULES COINCIDE. LOWER EPR IS BETTER EVERYWHERE.

| Name | Compiler | Layout source | $\|\mathcal{L}\|$ | $P$ | routes | selects on |
|---|---|---|---|---|---|---|
| *Own protocol* (RQ1, RQ4) | | | | | | |
| TELESABRE | TELESABRE | Hungarian, seeds 0–2 | 3 | 1 | 3 | best EPR |
| DSABRE | DSABRE | SabreLayout, seeds 0–2, both corner rules | 6[a] | 3 | 9–18 | best EPR |
| PORTER | PORTER | temporal locality-greedy | 1 | 3 | 3 | best EPR |
| PORTER+R | PORTER | temporal locality-greedy | 1 | 5 | 5 | best forward pass |
| PORTER+S | PORTER | $\gamma \in \{0.90, 0.95, 0.99, 1.0\}$ | 4 | 1–3 | 9–13 | best EPR |
| PORTER+SR | PORTER | the above two, pooled | 4 | — | 11–15 | best EPR |
| *Shared layout* (RQ2) — both compilers routed from one external layout | | | | | | |
| DSABRE shared | DSABRE | TELESABRE's or pytket-dqc's | 1 | 3 | 3 | best pass |
| PORTER shared | PORTER | the identical layout | 1 | 3 | 3 | best EPR |
| PORTER+R shared | PORTER | the identical layout | 1 | 5 | 5 | best forward pass |
| *Cross layout* (RQ2) — each compiler on the other's own layout | | | | | | |
| DSABRE on PORTER | DSABRE | PORTER's layout | 1 | 3 | 3 | best pass |
| PORTER on DSABRE | PORTER | DSABRE's candidates | 6[a] | 3 | 9–18 | best EPR |
| *Ablation* (RQ3) — one fixed layout, one fixed horizon | | | | | | |
| ablation legs | PORTER | temporal locality-greedy | 1 | 1 | 1 | — |

TABLE II

MAIN COMPARISON, OWN PROTOCOL: EACH COMPILER FROM ITS OWN INITIAL LAYOUT AND ROUTE BUDGET (TABLE I). PORTER+R IS PORTER WITH BACKWARD REFINEMENT. EPR COLUMNS ARE GEOMETRIC MEANS OVER THE CIRCUITS *all four* COMPILERS COMPLETE, SO THEY EXCLUDE THE SEVEN TELESABRE DOES NOT CONVERGE ON; "DONE" COUNTS THE CIRCUITS EACH COMPILER COMPLETES, OUT OF THE SUITE SIZE. $t$ IS COMPLETE PROTOCOL TIME (GEOMETRIC MEAN, SECONDS) OVER THE WHOLE SUITE. LOWER IS BETTER THROUGHOUT. PER-CIRCUIT EPR AND SWAP COUNTS ARE IN APPENDIX H.

| | TELESABRE | | DSABRE | | PORTER | | PORTER+R | |
|---|---|---|---|---|---|---|---|---|
| Suite | EPR | $t$ | EPR | $t$ | EPR | $t$ | EPR | $t$ |
| 25q (6) | 106.6 | 0.51 | 44.0 | 0.44 | 43.1 | **0.11** | **38.8** | 0.28 |
| 64q (9) | 284.5 | 11.89 | **144.6** | 1.86 | 166.2 | 1.98 | 149.2 | **1.85** |
| 80q (9) | 121.0 | 5.11 | **36.5** | 8.14 | 38.5 | **3.95** | 44.5 | 4.97 |
| done | 17 / 24 | | 24 / 24 | | 24 / 24 | | 24 / 24 | |

### H. Metrics and Aggregation

The primary metric is EPR pairs consumed. TELESABRE counts teledata plus telegate operations, DSABRE counts teledata, and PORTER counts teleports plus telegates plus telegate packets, one pair per packet however many remote CX gates it amortises; all three count the same physical resource. Intra-core SWAP counts and compile time are secondary and are never traded against EPR by any selection rule.

**Timing.** All compile times are wall-clock on one otherwise idle machine (Apple M2, macOS 26.5, CPython 3.13), measured once per configuration rather than as a median of repetitions, with no warm-up: each invocation is a fresh process, so no cache is carried between circuits. Two consequences follow. Times are indicative rather than precise, and should be read at the resolution the paper uses them — as ratios between protocols, not as absolute costs. And because horizon selection guards its speculative routes by wall-clock (Section IV-F1), machine load can in principle change which route it selects; the horizon each reported run chose is recorded with its result.

Aggregates are unweighted geometric means over circuits, $\left(\prod_i x_i\right)^{1/m}$, so each circuit counts equally and a suite with large absolute counts does not dominate. Percentage changes are $100(b/a - 1)$ against the stated baseline $a$; ratios are $b/a$. Both are computed from the geometric means, not averaged per circuit.

Failures are never imputed. A configuration that does not complete an instance is marked and excluded from every geometric mean it appears in, along with that instance's other columns where the comparison is paired. A mean labelled $(n)$ is therefore over exactly the $n$ circuits all its columns completed. This separates quality from completion, but it also means a baseline that fails its hardest instances is scored only on the ones it survives, which flatters it. We report completion counts alongside every such mean.

## V. RESULTS

Each subsection answers one research question of Section IV-A. Protocol names and budgets are those of Table I; lower EPR is better throughout.

### A. RQ1: End-to-End Cost Under Each Compiler's Own Protocol

*Question:* how do the three compilers compare when each uses its own recommended protocol? *Held fixed:* nothing — each compiler supplies its own layouts and pass schedule (Table I). *Metric:* geometric-mean EPR over jointly completed circuits, plus completion counts and complete compile time.

Table II summarises the three suites; Appendix H gives the per-circuit EPR and SWAP counts. Baseline counts are taken from each baseline's own harness and are not covered by PORTER's replay-verification guarantee (Section IV-C). Against TELESABRE the ordering is consistent across the suite. From one deterministic layout PORTER lowers geometric-mean EPR by 59.6%, 41.6% and 68.2% at 25, 64 and 80 qubits, spending three routes against TELESABRE's three seeded runs. Every circuit TELESABRE completes improves, from $-16.0\%$ to $-81.0\%$. Refinement adds to this at 25 and 64 qubits (63.6% and 47.6%) but not at 80, where the selected route is already stronger (68.2% against 63.2%) — the first place where choosing the horizon beats spending four more routes. PORTER and DSABRE complete all twenty-four instances; TELESABRE completes seventeen, failing on 64q Random and six of the nine 80-qubit circuits, including the 23,381-CX Random.

Two details are worth recording. On all eight 64-qubit circuits TELESABRE completes, our harness reproduces the EPR and SWAP counts DSABRE publishes for it [11] exactly, so the two evaluations share a baseline measurement rather than only a baseline tool. And GHZ, the chain circuit that defeats locally-greedy schedulers, costs TELESABRE 16 pairs on 63 CX gates but PORTER and DSABRE 6 each: PORTER's first-appearance visit order puts successive chain neighbours in one core before routing begins.

Against DSABRE the answer depends on the pass budget. At its own three-route protocol PORTER takes the geometric mean at 25 qubits (43.1 against 44.0) and trails at 64 and 80 (166.2 against 144.6; 38.5 against 36.5, both on the TELESABRE-matched basis), winning eight instances, losing twelve, tying four. Refinement comes within 3.2% at 64; the $\gamma$-search closes the 64-qubit suite outright (172.7 against 173.1

TABLE III
SHARED-LAYOUT COMPARISON: DSABRE AND PORTER BOTH ROUTED FROM THE IDENTICAL LAYOUT TELESABRE BUILT FOR ITSELF, SO THE ONLY DIFFERENCE LEFT IS THE ROUTING LOOP. PORTER+R IS PORTER WITH BACKWARD REFINEMENT, FIVE ROUTES AGAINST DSABRE'S THREE. GEOMETRIC-MEAN EPR OVER THE CIRCUITS TELESABRE COMPLETES AND THEREFORE PROVIDES A LAYOUT FOR — ALL SIX AT 25Q, EIGHT OF NINE AT 64Q, THREE OF NINE AT 80Q. $\Delta_{\mathrm{TS}}$ COLUMNS ARE EACH COMPILER'S CHANGE AGAINST TELESABRE'S OWN-PROTOCOL RESULT. PER-CIRCUIT ROWS ARE IN APPENDIX H.

| | | DSABRE | | PORTER | |
|---|---|---|---|---|---|
| Suite | TELESABRE | EPR | $\Delta_{\mathrm{TS}}$ | +R | $\Delta_{\mathrm{TS}}$ |
| 25q (6) | 106.6 | 51.0 | −52.2 | **47.7** | **−55.2** |
| 64q (8) | 284.5 | 189.2 | −33.5 | **157.2** | **−44.7** |
| 80q (3) | 121.0 | 74.2 | −38.7 | **55.7** | **−53.9** |
| **all** (17) | 173.0 | 100.9 | −41.7 | **86.0** | **−50.3** |

over all nine) and leaves 1.08× at 80. Its widest win is 64q Multiplier, 763 against 1,285 with refinement; its worst losses are 1.5×, on 80q QPEexact (526 against 347) and 64q QFT (262 against 179). The first is a loss for refinement alone — the selected route takes that instance at 291.

### B. RQ2: Routing Quality With the Layout Controlled

*Question:* with the initial layout held identical, does either routing loop use fewer EPR pairs? *Held fixed:* the initial layout, supplied externally. We report two comparisons: an *equal-route* one, three routes each, which is the controlled test; and a *refinement-schedule* one, DSABRE's three passes against PORTER+R's five, which is not budget-matched and does not isolate the router. *Metric:* geometric-mean EPR and win/loss counts.

The own-protocol comparison mixes two effects: the quality of each compiler's initial layout and the quality of its routing loop. Table III removes the first — both DSABRE and PORTER route from the one identical layout TELESABRE built for itself — leaving a pure routing comparison.

**Equal route count.** At three routes each, DSABRE leads at 25 and 80 qubits and PORTER at 64 (−52.2% against −41.5%, −38.7% against −33.1%, −33.5% against −36.5%), pooling to 100.9 against 107.7 over seventeen instances. This is the controlled test, and PORTER loses it: the two routers are close, but with the layout fixed and the route count equal DSABRE has the lower pooled count. Spending those three routes on lookahead horizons is not what closes the gap.

**Refinement schedules.** Spending five routes on the forward–backward–forward schedule does close it: PORTER+R reaches 86.0 against DSABRE's 100.9 and leads at every scale. This is not a budget-matched comparison — it is five PORTER routes against three DSABRE ones — so it does not isolate the router. What it shows is that the additional routing effort, and not the commitment rule, is what reverses the ordering here; Section V-C confirms that from the other side, since removing commitment moves EPR by 0.6% in PORTER's favour. (Table III reports the refined column; the equal-route one is in Appendix H.)

**Pooled fixed-layout grid.** A second source of externally supplied layouts — `pytket-dqc`'s hypergraph placements, routed by both compilers on the 25q and 64q suites (Appendix B; Appendix C offers that partition to PORTER as an extra layout candidate instead) — pools with Table III to give forty-one instances on which DSABRE and PORTER routed the *identical* layout. At three routes each, DSABRE wins twenty to sixteen with five ties, pooled geometric mean 134.2 against 139.0. With PORTER+R's five routes against DSABRE's three, PORTER wins twenty to fifteen with six ties, 126.6 against 134.2. The two routers are therefore close on a fixed layout, and which one leads is decided by how many routes each is allowed rather than by the routing loop itself.

**Each compiler on the other's layout.** Closing the remaining cell — routing each system from its own and from its counterpart's layout (Appendix H) — gives the same reading from the other side. Down a column the compilers are close: on DSABRE's layout, DSABRE reaches 163.9 and PORTER+R 161.1; on PORTER's, 171.2 and 171.4. Across a row the layouts are not: DSABRE's candidate set is the better start for *both*, by 4.3% for its own router and 6.0% for PORTER's. The best cell is PORTER+R on DSABRE's layout.

At these sizes, then, the two routing stages are worth about the same given the same start, and what separates them end-to-end is that DSABRE searches up to six candidate layouts where PORTER builds one — a deliberate trade that Section V-F prices. It does not hold at every size: from 150 physical qubits upward the same control becomes counterproductive and a different parameter binds (Section V-D).

### C. RQ3: Mechanism Ablation

*Question:* what does each mechanism contribute on its own? *Held fixed:* the initial layout, the lookahead horizon and every other parameter; exactly one mechanism is disabled per column. *Metric:* EPR, completion, and — for plan commitment, whose claim is about cost — compile time.

Table IV attributes PORTER's EPR consumption to its mechanisms by disabling each in isolation, holding the initial layout and every other parameter fixed; the per-action counts behind these rows are tabulated in Appendix G (the six omitted rows — GHZ and Graphstate at every scale — form no packets, never trigger relief and issue no telegates, so all configurations coincide there). Two columns go further than a per-primitive ablation: "Hop-wise" replaces PORTER's unit of commitment with DSABRE's while holding scoring, plan set and relief tiers fixed, and "TELEDATA" removes the TELEGATE family outright, leaving state teleportation as the sole inter-core action.

**Packets.** Disabling TELEGATEPACKET makes every shared-control remote CX run pay per gate, and the cost tracks the packet count: +42% EPR on 64q QFT (262 → 372, 45 packets), +40% on 80q AE (490 → 688, 264 packets), +19% on 64q AE, against a +7% overall gmean; the lone counter-move (80q QFT, 446 → 435) shows packet formation is a heuristic commitment that can displace a slightly better teleport plan. Read alone, this column is a clean win for the primitive. The next one overturns that reading.

**TELEDATA only.** This column is two configurations and must be read as such. *Strict removal* — deleting the TELE-

TABLE IV
MECHANISM ABLATION: PORTER EPR WITH ONE DESIGN DECISION DISABLED AT A TIME, ALL FROM THE SAME TEMPORAL LOCALITY-GREEDY LAYOUT. COLUMNS: "NO PK" DISABLES PACKETS (MAX_PACKET_SIZE=1), "NO REL." DISABLES BOTH RELIEF TIERS, "HOP-WISE" REPLACES PLAN COMMITMENT WITH PER-ITERATION RE-SCORING OF THE WHOLE FRONT, "TELEDATA" REMOVES THE TELEGATE FAMILY; "PK" AND "TG" ARE THE PACKET AND SINGLE-TELEGATE COUNTS OF THE FULL CONFIGURATION. EACH GMEAN ROW IS OVER THE INSTANCES *all its filled columns* COMPLETE, AND ONLY THOSE COLUMNS ARE FILLED: $n$=14 PAIRS FULL AGAINST NO PK AND NO REL., $n$=16 PAIRS FULL AGAINST TELEDATA, $n$=12 PAIRS FULL AGAINST HOP-WISE. ROWS ARE THEREFORE COMPARABLE ACROSS A ROW AND NOT DOWN A COLUMN. †FAILS TO COMPLETE (DEADLOCK OR ITERATION-GUARD TRIP). ‡COMPLETES ONLY WITH THE ADDITIONAL RELIEF TIER DESCRIBED IN THE TEXT, WHICH THE TELEGATE OTHERWISE MAKES UNNECESSARY. THE FIRST GMEAN ROW IS OVER THE FOURTEEN INSTANCES THE FIRST THREE CONFIGURATIONS SHARE; THE SECOND IS THE TELEDATA-ONLY COLUMN PAIRED AGAINST FULL OVER THE SIXTEEN INSTANCES IT COMPLETES.

| Suite | Circuit | Full | No pk | No rel. | Hop-wise | TELEDATA | pk | tg |
|---|---|---|---|---|---|---|---|---|
| 25q | ae | 65 | 71 | 80 | 65 | 68‡ | 8 | 0 |
| 25q | qft | 74 | 74 | 140 | 74 | 74 | 0 | 0 |
| 25q | qnn | 119 | 122 | 212 | 128 | 119 | 0 | 0 |
| 25q | random | 401 | 412 | 395 | 399 | 413‡ | 43 | 93 |
| 64q | ae | 177 | 211 | 327 | 172 | 198‡ | 32 | 0 |
| 64q | qft | 262 | 372 | 382 | 336 | **216**‡ | 45 | 1 |
| 64q | qnn | 864 | 931 | 600 | 785 | **694**‡ | 240 | 5 |
| 64q | random | 737 | 843 | 789 | 805 | † | 77 | 228 |
| 80q | ae | 490 | 688 | † | 383 | **354**‡ | 264 | 15 |
| 80q | qft | 446 | 435 | † | 415 | **307**‡ | 209 | 2 |
| 80q | qnn | 1517 | 1571 | † | 1474 | **926**‡ | 448 | 30 |
| 80q | random | 11040 | 11628 | † | 11280 | † | 572 | 3538 |
| **gmean**, paired $n$=14 | | 57.6 | 61.4 | 67.0 | | | | |
| **gmean**, paired $n$=16 | | 78.3 | 83.9 | | | **71.7** | | |
| **gmean**, paired $n$=12 | | 413.3 | | | 411.7 | | | |

GATE family and changing nothing else — is the one-factor ablation, and it fails: ten of the twelve rows do not complete. The TELEGATE is the only remote action needing no free destination slot, so it is always a legal plan for an adjacent-remote CX; without it, a front whose remote CXs are all core-adjacent and whose destinations sit at or below $f_{\min}$ has an empty plan set, while the tail-relief BFS is offered only the non-adjacent blocked candidates. The TELEGATE is thus essential to successful *completion* here under the unchanged relief policy — not to termination, which the guards of Section III-H provide either way.

*The vocabulary variant* adds one mechanism the full configuration never needs: adjacent-blocked candidates are fed into the same relief BFS. Every entry marked ‡ uses it. It recovers all but the two Random rows, which livelock — and those two are the most telegate-heavy in the suite (228 and 3,538 single telegates). As a control, adding the same tier to the full configuration leaves every row EPR-identical, so the comparison below is a difference of vocabulary rather than of relief. It is not, however, a strict ablation.

On that basis the variant is cheaper than the full action set, increasingly so with scale: $+2\%$ at 25q, mixed at 64q, $-33\%$ at 80q, and $8.5\%$ below Full over the sixteen instances it completes. The circuits it wins are the packet-heaviest, so the ordering there is TELEDATA only $<$ Full $<$ No packets: packets beat the single telegates they degrade into, but a pure teleportation policy beats both. Section VI argues this is a currency mismatch in the plan comparator rather than a verdict on packets. It also answers a question DSABRE leaves open. Its teledata-only vocabulary was chosen by measurement — an ungrouped telegate serves one gate where teledata amortises many — with grouping untested. Grouping does close that per-EPR gap, and still does not justify spending pairs that teleporting would have spent better.

**Plan commitment.** Replacing the commitment rule with DSABRE's — returning to the top of the loop after one round of preparation SWAPs, re-classifying the front and re-scoring every plan before the next move — leaves EPR unchanged: 413.3 against 411.7 over the twelve rows here, and 148.8 against 148.0 over the twenty-one instances we ran it on, with eight circuits better without commitment, five worse, eight tied, and all twenty-one completing either way. The two make the same number of routing moves (128,222 against 128,565 iterations), which is the key number: commitment does not change what the compiler decides, only what each decision costs. It buys $1.31\times$ compile time in geometric mean and $3.07\times$ on 80q Random. Section VI notes what this does and does not say about the plan abstraction beneath it.

**Capacity relief.** Disabling both tiers costs completion, not just EPR pairs: 80q AE, QFT, QNN and Random all fail, because with every destination below $f_{\min}$ and no way to drain qubits toward reservoirs the loop has no legal remote plan left. Where routing does complete, dense circuits pay heavily ($+89\%$ on 25q QFT, $+78\%$ on 25q QNN, $+85\%$ on 64q AE). Two instances are cheaper *without* relief — 64q QNN ($864 \to 600$) and marginally 25q Random ($401 \to 395$) — showing the opportunistic tier can overpay where congestion would have cleared itself. Relief is thus the one mechanism here whose removal costs completion: with both tiers on, no instance in this evaluation reached the deadlock path of Section III-H; with both off, four did. That is a statement about these instances, not a guarantee. The initial layout, which this table holds fixed, is assessed in Sections V-B and V-E.

### *D. RQ4: Scale-Up on QFT and QPEexact to 360 Logical Qubits*

*Question:* do the RQ1 and RQ2 answers hold as circuit width and core count grow together? *Held fixed:* DSABRE's own scalability architectures and circuit files. *Metric:* geometric-mean EPR and complete compile time.

Every result so far is at 108 physical qubits or fewer. Table V takes the two circuit families that exist at 100, 200 and 360 logical qubits onto DSABRE's scalability series: $5\times5$ cores throughout, core count $6 \to 12 \to 20$, so occupancy stays roughly constant while the core graph grows. It is a monotone ladder in the quantity that matters — inter-core traffic scales with core count, and DSABRE's own cost rises $200 \to 765 \to 1{,}890$ on QFT.

PORTER uses fewer EPR pairs than DSABRE on all six instances, by a widening margin: $0.90\times$ at 100 logical qubits, $0.67\times$ at 200, $0.62\times$ at 360, for $0.72\times$ overall. It does so at $0.20\times$ DSABRE's complete compile cost (8.4 s against 41.1 s), because DSABRE routes up to eighteen times here against PORTER's three.

TABLE V

SCALE-UP SUITE: QFT AND QPEEXACT AT 100, 200, AND 360 LOGICAL QUBITS, EPR PAIRS. ARCHITECTURES FOLLOW DSABRE'S OWN SCALABILITY PROTOCOL — 5×5 CORES THROUGHOUT, CORE COUNT $6 \to 12 \to 20$ — READ FROM THE BASELINES' OWN DEVICE FILES SO ALL ROUTERS SEE IDENTICAL PHYSICAL QUBIT IDS; OUR DSABRE COLUMN REPRODUCES THE FIGURES IT PUBLISHES FOR EXACTLY THIS SERIES [11] (201/246/766/896/1890/1938) TO WITHIN THE ONE CANDIDATE LAYOUT OUR PROTOCOL ADDS. THE LAST TWO COLUMNS ARE THE FIXED-LAYOUT CONTROL: PORTER ROUTING DSABRE'S OWN SABRELAYOUT CANDIDATES. "$t$" IS COMPLETE PROTOCOL TIME IN SECONDS — LAYOUT GENERATION PLUS EVERY ROUTE OVER EVERY CANDIDATE, THE METRIC USED THROUGHOUT: NINE TO EIGHTEEN ROUTES FOR DSABRE, THREE FOR PORTER. PYTKET−DQC IS OMITTED: ITS HYPERGRAPH PARTITIONER DOES NOT FINISH IN REASONABLE TIME AT THESE WIDTHS. LOWER IS BETTER IN EVERY COLUMN.

| | | architecture | | | DSABRE | | PORTER, own layout | | | PORTER, DSABRE layout | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Circuit | CX | $n$ | $P$ | $K$ | EPR | $t$ | EPR | $t$ | +R | EPR | +R |
| QFT | 3,420 | 100 | 150 | 6 | 200 | 11.6 | 198 | 2.8 | 339 | 240 | 273 |
| QPEexact | 3,579 | 100 | 150 | 6 | 246 | 13.0 | 200 | 2.2 | 262 | 200 | 283 |
| QFT | 7,220 | 200 | 300 | 12 | 765 | 47.9 | 579 | 9.5 | 1128 | 1031 | 1018 |
| QPEexact | 7,579 | 200 | 300 | 12 | 896 | 49.5 | 528 | 8.9 | 979 | 651 | 946 |
| QFT | 13,300 | 360 | 500 | 20 | 1890 | 93.0 | 1156 | 19.5 | 2822 | 1870 | 2546 |
| QPEexact | 13,979 | 360 | 500 | 20 | 1938 | 143.8 | 1224 | 33.1 | 2342 | 1444 | 2416 |
| **gmean** (6) | | | | | **705.7** | **41.1** | **507.7** | **8.4** | 930.3 | 665.7 | 877.9 |

**Of the interventions we tested, only the horizon explains the reversal.** Both mechanisms that close PORTER's gap at the smaller sizes are inert or harmful here. Backward refinement loses on all six instances, by 83%. Handing PORTER DSABRE's own `SabreLayout` candidates — the control that reaches parity at 25 and 64 qubits and wins on both heavy-hex core graphs — makes it *worse*, 0.94× against 0.72×, and worst on the largest instance, where DSABRE's layout costs 1,870 pairs against 1,156 from PORTER's own. Every instance in this series selects $L_{\text{tp}}$=128, and at the configured 16 PORTER trails DSABRE by 1.45× (Section IV-F1). We do not claim no other mechanism matters, only that among those we varied the horizon is the one that accounts for the change.

This separates two deficits the smaller suites conflate. At 36 to 108 physical qubits the initial layout limits PORTER: control it, match the route count, and the difference largely disappears. At 150, 300 and 500 physical qubits it does not, and the horizon does. Selecting it per circuit is a workaround, not a fix; a $K$-aware horizon remains open.

### E. RQ4: Heavy-Hex Cores and Non-Grid Core Graphs

*Question:* how much of the foregoing depends on square cores wired in a grid? *Held fixed:* the 64-qubit circuit suite and the physical-qubit count; the core shape and core graph change. *Metric:* geometric-mean EPR and win/loss counts.

Every result so far uses square cores wired in a grid. To test how much of PORTER's behaviour depends on that we adopt DSABRE's non-grid experiment [11]: the same nine-circuit 64-qubit suite on four IBM 27-qubit heavy-hex tiles. Two things change at once. The *core* becomes irregular — degrees over $\{1, 2, 3\}$, six degree-1 leaves, diameter 12 against a 4×4 grid's 6 — and the *core graph* is either a ring, where every core has degree 2, or a star, where core 0 is a hub. PORTER's hardware object is derived from the architecture DSABRE's own builder produces, and a topology assertion checks that both compilers see identical qubit ids, intra-core edges and port endpoints before anything is routed. DSABRE routes on its native object and reproduces its published numbers on all eighteen instances. Neither compiler is specialised for a heavy-hex tile.

TABLE VI

THE NINE-CIRCUIT 64-QUBIT SUITE ON FOUR IBM 27-QUBIT HEAVY-HEX CORES (108 PHYSICAL QUBITS), WIRED AS A *ring* (4 LINKS, EVERY CORE DEGREE 2) AND A *star* (3 LINKS, CORE 0 A HUB EVERY ROUTE MUST CROSS). GEOMETRIC-MEAN EPR; THE RIGHT-HAND PAIR IS THE FIXED-LAYOUT CONTROL, PORTER ROUTING DSABRE'S OWN SABRELAYOUT CANDIDATES. PORTER COMPLETES ALL EIGHTEEN INSTANCES, SO EVERY ROW IS OVER THE FULL NINE; PORTER+R, WHICH DOES NOT SELECT A LOOKAHEAD HORIZON, STILL FAILS STAR MULTIPLIER AND IS THEREFORE "—" ON THAT ROW. PER-CIRCUIT ROWS ARE IN APPENDIX H. DSABRE REPRODUCES ITS PUBLISHED NUMBERS FOR THESE TWO ARCHITECTURES ON ALL EIGHTEEN INSTANCES [11].

| | | own layout | | DSABRE layout | |
|---|---|---|---|---|---|
| Core graph | DSABRE | PORTER | +R | PORTER | +R |
| ring (9) | 85.2 | 86.1 | 82.3 | 82.9 | **81.4** |
| star (9) | 101.1 | 122.3 | — | 98.5 | **93.5** |
| star Multiplier | 722 | **455** | — | 462 | 462 |

**The core graph matters more than the core shape.** Table VI shows that replacing square-grid cores with these heavy-hex tiles costs PORTER nothing on the ring: on the ring PORTER+R leads DSABRE (82.3 against 85.2), winning six of nine and tying GHZ — better relative standing than on the grid suite. Doubling the intra-core diameter and replacing degree-4 interiors with degree-1 leaves disturbs neither compiler. The star does. Under its own protocol PORTER trails there, 122.3 against 101.1, winning two of nine; PORTER+R, which cannot route Multiplier at all, trails by 13.6% on the eight it completes.

**Star Multiplier is a livelock the horizon escapes.** At a fixed $L_{\text{tp}}$=16 this instance does not terminate, and it fails as a local-SWAP livelock rather than the capacity deadlock of Section III-H: retired gates freeze at 3,226 of 13,040 and the EPR count at 538 while intra-core SWAPs accumulate at one per iteration, so the iteration guard rather than the progress argument stops it. The failure is configuration-dependent rather than a property of the circuit alone: the same circuit, architecture, layout and router complete at $L_{\text{tp}}$=128, at 455 pairs against DSABRE's 722, and livelock at $L_{\text{tp}}$=16. Since horizon selection routes that candidate as a matter of course, PORTER has no unrouted instance anywhere in this evaluation — whereas PORTER+R, which holds the horizon fixed, still fails it.

**The initial layout is the dominant observed source of the star gap.** Given DSABRE's layout, PORTER completes all nine star circuits, including Multiplier at 462 pairs against DSABRE's 722. With the budget matched, that control reverses the result on *both* core graphs (−4.4% ring, −7.5% star), the larger margin on the topology PORTER loses under its own protocol. The cause is in the layout: PORTER reserves $f_{\min}$ in every core, treating cores as interchangeable. They are on a grid and a ring, where every core has the same degree. They are not on a star, where Multiplier opens with free counts $(2, 2, 13, 27)$ — the hub every route must cross pinned at the

TABLE VII
PROTOCOL TRADE-OFF OVER ALL 24 INSTANCES. "ROUTES" IS THE NUMBER OF ROUTING PASSES; TIMES ARE COMPLETE COMPILE COSTS (GEOMETRIC MEAN PER INSTANCE). PORTER ROUTES EACH OF THREE CANDIDATE LOOKAHEAD HORIZONS AND KEEPS THE BEST (SECTION IV-F1); PORTER+R ADDS BACKWARD REFINEMENT (SECTION IV-F2) AT A FIXED HORIZON; PORTER+S ADDS THE BEST-OF-$\gamma$ LAYOUT SEARCH OF THIS SECTION ON TOP OF IT; PORTER+SR TAKES THE BETTER OF HORIZON SELECTION AND THAT SEARCH, NETTING THE ONE ROUTE THEY SHARE.

| Protocol | routes | EPR | vs. DSABRE | time (s) | vs. DSABRE |
|---|---|---|---|---|---|
| DSABRE | 9–18 | 163.9 | 1.00 | 2.26 | 1.00 |
| PORTER | 3 | 174.9 | 1.07 | 1.32 | **0.58** |
| PORTER+R | 5 | 171.4 | 1.05 | 1.67 | 0.74 |
| PORTER+S | 9–13 | 163.2 | 1.00 | 4.59 | 2.03 |
| PORTER+SR | 11–15 | **158.2** | **0.97** | 5.65 | 2.50 |

floor while a whole core sits empty. Raising the reservation uniformly does not repair it (we tried $f_{\min} \in \{4, 6\}$); what the star needs is a reservation scaled by a core's position in the core graph, which we have not implemented.

Taken with Section V-D, this bounds the layout reading rather than extending it. Up to 108 physical qubits — across two core shapes and three core graphs — the initial layout separates the two compilers and controlling it erases the difference. Above 150 the horizon takes over.

### F. The EPR/Time Operating Frontier

Table VII places PORTER's four operating points (Section IV-F) against DSABRE, pooled over all twenty-four instances. Selection and the $\gamma$-search look for different things — the horizon at a fixed layout against the layout at a fixed horizon — and neither dominates: the search gives the better route on fourteen instances, selection on four, and they tie on six. PORTER+SR runs both.

The trade is time against EPR pairs and it is monotone in both. PORTER costs $1.07\times$ DSABRE's EPR at $0.58\times$ its compile time; refinement reaches $1.05\times$ at $0.74\times$; the layout search reaches parity (163.2 against 163.9) at $2.03\times$; PORTER+SR passes it, at $0.97\times$ its EPR for $2.50\times$ its time. PORTER+SR leads DSABRE on twelve instances against nine behind, and by construction never loses to either schedule it is built from. Increasing PORTER's deterministic search budget therefore closes and reverses the own-protocol gap. We stop short of calling that gap a search-budget gap: the added schedules change the initial layout as well as the route, and the comparison equalises route counts rather than wall-clock time. Where to sit on the curve depends on whether EPR pairs or compile seconds are scarce.

Selecting the horizon has largely overtaken refinement as an operating point. PORTER+R is ahead by only $2.0\%$ in geometric mean for four extra routes, and it is $12.0\%$ behind the selected route at 80 qubits, $83\%$ behind on the scale-up series, and the configuration that fails star Multiplier. We report it because the fixed-layout comparison of Section V-B needs a schedule that spends its extra effort on routes rather than layouts, which PORTER+S cannot be.

All times are *complete* compile costs: layout generation plus every route over every candidate, not the winning route alone. That is nine to eighteen routes for DSABRE, three for PORTER, five for PORTER+R, and all three seeds for TELESABRE. PORTER's advantage is in the count of routes, not their price. It is the dearer of the two per route on all nine 64-qubit circuits, taking $3.7\times$ as long in geometric mean: enumerating whole plans, collecting packets by forward simulation and running a two-tier capacity search all cost more per iteration than scoring a flat list of hops, and a deep horizon costs more still. On that subset the complete costs are 1.98 s against 1.86 s, so DSABRE is cheaper; pooled over all twenty-four PORTER leads at $0.58\times$.

Over the five circuits TELESABRE completes, geometric-mean complete cost is 5.95 s for TELESABRE, 2.67 s for DSABRE, 1.55 s for PORTER and 1.97 s with refinement (per-circuit times in Appendix H). Neither PORTER point holds that margin on the largest circuit, where PORTER+R costs 191 s against 144 s at 23,381 CX. The cheap operating point is still PORTER, and at $1.07\times$ DSABRE's EPR it gives up little to be so.

## VI. DISCUSSION AND LIMITATIONS

Several results in Section V read as tensions when placed side by side. We resolve them here, then state what the evaluation does not establish.

**The plan abstraction and the commitment rule are separable.** The *plan-oriented* design — whole inter-core actions as candidates, ranked on gain with preparation cost demoted to a tie-break — is the defining structural difference in PORTER's remote stage. *Commitment* is the narrower rule that a selected plan is not reopened until ready. Only the latter is ablated: the hop-wise column re-derives and re-scores the same plan set every iteration, changing when the choice is revisited and nothing else. That it reaches the same EPR on the same number of moves says commitment is not where quality comes from. It says nothing either way about the abstraction, which no column removes. We therefore do not attribute the reductions of Section V-A to it: those are produced by the whole system — layout, relief, vocabulary, scoring, horizon and protocol together — and the present ablations do not isolate the abstraction's share. Doing so would need a hop-granular candidate set, which is a different router. This is the clearest gap in our attribution.

**Layout binds below about a hundred physical qubits, the horizon above it.** PORTER trails DSABRE under RQ2 at $\leq 108$ physical qubits, and controlling the layout at a matched budget erases the difference. On the scale-up series neither that control nor extra refinement helps — DSABRE's layout is worse for PORTER there than its own — while a longer horizon moves PORTER from $1.45\times$ behind to $0.72\times$ ahead. Both regimes are consistent with the horizon being an absolute gate count that does not grow with the core count $K$, so a plan at twenty cores is scored over proportionally less of the circuit than at four. We have not tested a $K$-scaled horizon, so this is a hypothesis, not a finding.

**Why TELEDATA-only can beat the full action set.** Packets beat the single telegates they degrade into, yet removing gate teleportation entirely is cheaper still on the packet-heaviest 80-qubit circuits. The cause is a currency mismatch in the scorer.

A packet's gain counts remote CX gates resolved immediately; a TELEDATA plan's is a sum of core-distance reductions, most of it discounted over a bounded horizon. Nothing establishes that one resolved gate and one unit of distance reduction are commensurate, and where a teleport repays past the horizon the comparator cannot see it. The ablation is evidence that this mismatch costs EPR pairs at scale; it is not evidence that a pair spread over a run of gates is a bad idea.

**What the completion record does and does not show.** Section III-H separates correctness and termination, which PORTER guarantees by construction, from completion, which it does not. Every instance in this evaluation completes under the selected-horizon protocol, but that is a property of the protocol and the benchmark rather than of the router: at a fixed horizon, heavy-hex star Multiplier livelocks, and the extra candidate is a way of not meeting the failure rather than of excluding it. Two of the four failure modes of Section IV-B were observed here — livelock at a fixed horizon, and infeasibility on an external layout that fills a core — while capacity deadlock appeared only with relief disabled. DSABRE's safe mode carries a termination-with-success proof under stated architectural hypotheses, and supplying a comparable guarantee is the clearest piece of theory this design leaves open.

### A. Threats to Validity

**Benchmark coverage.** The main suites are nine MQT-Bench families at three widths; the scale-up series is two families (QFT and QPEexact) only, so nothing here supports a claim about scaling behaviour on irregular or randomly structured circuits at those widths. The two heavy-hex networks are two core graphs, not a survey of architectures.

**Parameter rules fitted in-sample.** The horizon candidate sets and the density threshold were chosen by inspecting these same instances (Section IV-F1). The magnitude of the horizon-selection gain is therefore in-sample. The portfolio construction bounds the downside — the configured horizon is always a candidate and wins ties — but an out-of-sample validation has not been performed. The same caution applies in weaker form to $f_{\min}$, $m_{\text{assist}}$ and the packet window, which were fixed early and not swept.

**Reconstructed benchmark instances.** The 80-qubit AE and QNN circuits are not produced by the stock generator and were reconstructed (Appendix A). We release the construction scripts and the files, and validate the QNN reconstruction by exact CNOT-sequence match against the 64-qubit file, but they remain our circuits rather than the generator's.

**Patched baseline binary.** All TELESABRE numbers use a binary patched to avoid an out-of-range index in its initial-layout pass (Section IV-D). The patch changes placement only for qubits the stock binary leaves unplaced, and architecture-H results, where the stock binary does not crash, are unchanged by it; we have not otherwise audited its effect.

**Unequal native budgets.** The own-protocol comparison gives DSABRE nine to eighteen routes against PORTER's three. This is each compiler's own convention rather than a controlled budget, which is why RQ2 exists and why compile time is reported everywhere.

**Aggregation over failures.** Geometric means are taken over jointly completed instances, so a baseline that fails hard instances is scored only on easier ones. Completion counts are reported alongside every such mean, but the two must be read together.

**Implementation and timing.** PORTER is Python and the baselines are Python and C respectively, so per-pass times compare implementations as much as algorithms. Compile times are wall-clock on one machine. Horizon selection additionally guards its speculative routes with a wall-clock budget, so on a loaded machine it may abandon a candidate that would otherwise have won: the protocol's output is reproducible only up to that budget, though each individual route is exactly reproducible.

**Cost model.** One EPR pair per inter-core action, independent of hop count and generation protocol; link contention, entanglement latency and fidelity, purification, and classical communication cost are all outside the model (Section II-B). Conclusions are relative to that model.

## VII. RELATED WORK

**TELESABRE** [10] is the published state of the art for this cost model: it supports teledata and gate teleportation and builds on SABRE's extended-set scoring. The two differ chiefly in decision structure. TELESABRE scores inter-core candidates through a contracted-communication-graph Dijkstra energy that folds staging cost and lookahead into one value, and — uniquely among the compilers here — places those candidates in the *same* pool as local SWAPs, taking the global minimum each iteration. PORTER keeps local work strictly ahead of inter-core work, scores whole plans independently of preparation cost, generates preparation SWAPs only for the plan it commits to, and adds the packet primitive. Its capacity handling is also more proactive: alongside TELESABRE's safety-valve cap it maintains an explicit blocked-plan set with a tail-relief search and a pre-emptive relief comparator.

**DSABRE** [11] attacks the same problem, and the two compilers are close relatives. We run it in its default configuration throughout, and our numbers match those it publishes for the architectures it reports.

Stating what the two share keeps either from being credited with the other's ideas: both rank moves by SABRE's front-plus-lookahead $\Delta$ score; both are strictly locality-first, scoring an inter-core candidate only once no front gate is intra-core; both promote capacity from a scoring penalty to a legality condition with the same floor of two free slots; and both manufacture capacity by relaying a free slot from a reservoir along a BFS path rather than evicting arbitrarily. Four differences remain, and they are what this paper evaluates: PORTER's action vocabulary is larger (DSABRE ships teledata only); its unit of commitment is a whole plan rather than one hop; preparation cost enters as a tiebreaker rather than additively; and its initial layout is one deterministic construction rather than a seeded best-of-$N$ search. DSABRE's safe mode additionally carries a completion guarantee that PORTER does not match (Sections III-H and VI).

TABLE VIII
DESIGN AXES OF THE COMPILERS DISCUSSED HERE. "COMMITMENT" IS THE UNIT A SCORING DECISION IS TAKEN OVER. `PYTKET-DQC` IS PARTITION-FIRST AND DOES NOT ROUTE WITH THESE PRIMITIVES, SO SEVERAL AXES DO NOT APPLY.

| | TELESABRE | DSABRE | pytket-dqc | PORTER |
|---|---|---|---|---|
| Paradigm | online | online | partition-first | online |
| Primitives | data, gate | data | (e-bit) | data, gate, packet |
| Local/remote | one pool | staged | — | staged |
| Commitment | one move | one hop | whole plan | whole plan |
| Prep. cost | in score | in score | — | tie-break |
| Layout | Hungarian, 3 seeds | SabreLayout, 3 seeds | hypergraph partition | one deterministic |
| Capacity | cap | legality + relay | partition | legality + relief |
| Termination | — | with-success proof | — | correct, terminates |
| Metric | EPR | EPR | e-bits | EPR |

Table VIII places the four systems on the axes that matter here.

**Other approaches.** Partition-first compilers assign qubits to cores by hypergraph or community partitioning [13]–[15] and route the residue; `pytket-dqc` [14] is the mature representative, compared against in Appendix B under the caveat that its e-bit unit assumes long-lived entanglement and different communication memory. AutoComm [8] identifies gate sequences that can share one entangled resource, which PORTER's packet applies inside a routing loop rather than as a pre-pass (Section II-C). SABRE [12] supplies the scoring skeleton all three compilers inherit.

## VIII. CONCLUSION

PORTER is a deterministic DQC compiler built on a plan-oriented remote stage: it scores a complete inter-core action on the gate progress it buys, before any preparation SWAP for it exists, and commits before preparing. Around that it places a capacity discipline with two relief tiers, a deterministic initial layout, and a per-circuit choice of teleport-lookahead horizon.

Three conclusions are supported. First, on the evaluated MQT-Bench suites PORTER uses substantially fewer EPR pairs than published TELESABRE — $59.6\%$, $41.6\%$ and $68.2\%$ less in geometric mean at 25, 64 and 80 logical qubits, at three routes each — and completes every instance in all three suites, including the seven on which TELESABRE does not converge. Second, against DSABRE, PORTER is competitive at the sizes both target — $1.07\times$ its pooled EPR at $0.58\times$ its compile time under each compiler's own protocol — and its advantage grows with the machine, reaching $0.72\times$ its EPR pairs on the 100–360-qubit scale-up series. At the smaller sizes which compiler leads depends on the initial layout and the routing budget, and Section V-B reports both directions. Third, the ablations attribute these results rather than leaving them to the system as a whole: capacity relief is what lets a deterministic-greedy design complete in practice; the initial layout dominates the difference between the two compilers at the smaller sizes evaluated, and the lookahead horizon at the larger ones.

Two ablation results are negative and they delimit the contribution. Plan commitment does not lower EPR consumption — it makes the same number of routing moves as re-scoring the front each iteration and lands within $0.6\%$ — so it is a compile-time mechanism, and no experiment here isolates the plan abstraction it rests on. And a richer gate-teleportation vocabulary does not always pay: at 80 qubits, removing gate teleportation entirely beats the full action set by $32.8\%$ on the packet-heaviest circuits, which we read as the plan scorer undervaluing teleportation's latent return rather than as a verdict on amortisation.

The evaluation is bounded: two circuit families at the largest widths, two non-grid core graphs, horizon rules fitted on the corpus they are measured on, and a cost model charging one pair per inter-core action that ignores link contention, latency and fidelity (Section VI-A). Five openings follow: a completion guarantee to match DSABRE's, an out-of-sample validation of the horizon rule, a horizon that scales with core count rather than being chosen by trial, a comparator that weighs a resolved gate against a latent distance reduction on one scale, and a hardware cost model under which EPR count is not the whole objective.

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